1 A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 1.1.

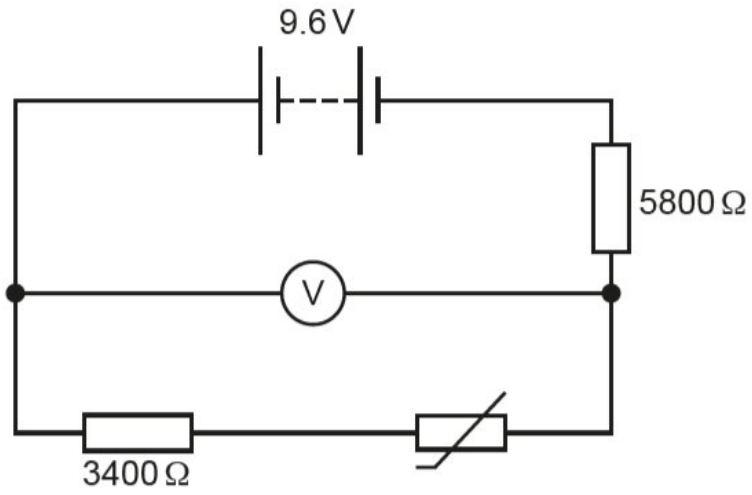


Fig. 1.1

The fixed resistors have resistances of 3400 Ω and 5800 Ω. The reading on the voltmeter in the circuit is 6.0 V.

(a) Calculate the current in the resistor of resistance 5800 Ω.

current = ... A [2]

(b) Calculate the resistance of the thermistor.

resistance = ... Ω [2]

(c) The initial energy stored in the battery is 2.6×10^4 J.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

final stored energy = ... J [2]

(d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

(i) the temperature of the thermistor

..... [1]

(ii) the current in the thermistor

..... [1]

(iii) the potential difference across the thermistor.

..... [1]

(e) If the temperature of the thermistor in Fig. 1.1 changes at a constant rate, state a reason why the potential difference across the thermistor does not change at a constant rate.

.....

.....

..... [1]

[Total: 10]